Abstract

We study the topological properties of phonons in trapped Rydberg atom arrays, which arise from dipole-dipole interactions between the atoms. For various one-dimensional geometries, from zigzag to armchair configurations, we analyze the symmetries of the phononic Hamiltonian which give rise to topologically localized phonon excitations on the system boundaries. Because the phonon-phonon interactions here do not naturally respect chiral symmetry, which is crucial for realizing 1D topological phases, we show how an appropriate, geometry-dependent choice of the dipole moment orientation can restore this symmetry and enable topological characterization. The interplay between two phononic degrees of freedom in the harmonic trap potential, mediated by the dipole-dipole interactions, leads to interesting topological phases with winding numbers between zero and two. The corresponding edge states exhibit strong localization and robustness, making them potentially useful for applications in Rydberg-array-based quantum transport.

As an outlook we aim to explore how the phonon topology influences transport phenomena. Topological states in the electronic degrees of freedom in Rydberg arrays have already been shown to support topologically protected photon pumping [1–5]. Studying the effect of spin-phonon coupling can provide new control mechanisms in the realm of topological transport.

Hamiltonian

I am considering the interaction term H_I of a Hamiltonian of a system of N particles with positions $\{\bar{\mathbf{R}}_l\}_{l\in\{1,\ldots,N\}}$. It shall consist of two-body interactions, which depend on the position of the involved particles. Thus H_I can be written in the form

$$H_I(\{\bar{\mathbf{R}}_l\}) = \sum_{i,j;\, i\neq j} V_{i,j}(\bar{\mathbf{R}}_i, \bar{\mathbf{R}}_j)$$

In order to being able to derive the second order expansion of H_I, that will be needed later, in a more formal way, I write the two-body interactions as dependend on all positions.

$$H_I(\{\bar{\mathbf{R}}_l\}) = \sum_{i,j;\, i\neq j} V_{i,j}(\{\bar{\mathbf{R}}_l\})$$

The $V_{i,j}$ are formally functions of all positions $\{\bar{\mathbf{R}}_l\}$, i.e $V_{i,j}(\{\bar{\mathbf{R}}_l\})$. But of course they only depend on $\bar{\mathbf{R}}_i$ and $\bar{\mathbf{R}}_j$, i.e.

$$\frac{\partial V_{i,j}}{\partial \bar{\mathbf{R}}_l} = 0 \quad \text{if } i \neq l \neq j$$

Now I expand the interaction Hamiltonian in small displacements $\{\delta\mathbf{R}_l\}$ from the tweezer centers $\{\mathbf{R}_l\}$.

$$\begin{aligned}
H_I(\{\mathbf{R}_l + \delta\mathbf{R}_l\}) &= H_I(\{\mathbf{R}_l\}) + \sum_n \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n}\,\delta\mathbf{R}_n + \frac{1}{2}\sum_{n,m} \delta\mathbf{R}_n\,\frac{\partial^2 H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n \partial \mathbf{R}_m}\,\delta\mathbf{R}_m \\
&= H_I(\{\mathbf{R}_l\}) + \sum_n \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n}\,\delta\mathbf{R}_n + \frac{1}{2}\sum_{n\neq m} \delta\mathbf{R}_n\,\frac{\partial^2 H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n \partial \mathbf{R}_m}\,\delta\mathbf{R}_m + \frac{1}{2}\sum_{n=m} \delta\mathbf{R}_n\,\frac{\partial^2 H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n \partial \mathbf{R}_m}\,\delta\mathbf{R}_m \\
&= H_I(\{\mathbf{R}_l\}) + \sum_n \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n}\,\delta\mathbf{R}_n + \frac{1}{2}\sum_{n\neq m}\sum_{i,j;\, i\neq j} \delta\mathbf{R}_n\,\frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n \partial \mathbf{R}_m}\,\delta\mathbf{R}_m \\
&+ \frac{1}{2}\sum_n \sum_{i,j;\, i\neq j} \delta\mathbf{R}_n\,\frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n^2}\,\delta\mathbf{R}_n \\
&= H_I(\{\mathbf{R}_l\}) + \sum_n \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n}\,\delta\mathbf{R}_n + \frac{1}{2}\sum_{n\neq m}\sum_{i,j;\, i\neq j} \delta\mathbf{R}_n\,\frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n \partial \mathbf{R}_m}\,\delta\mathbf{R}_m\,(\delta_{n=i,m=j} + \delta_{n=j,m=i}) \\
&+ \frac{1}{2}\sum_n \sum_{i,j;\, i\neq j} \delta\mathbf{R}_n\,\frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n^2}\,\delta\mathbf{R}_n\,(\delta_{n=i} + \delta_{n=j})
\end{aligned}$$

Using the fact, that

$$\delta\mathbf{R}_i \, \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial\mathbf{R}_i\partial\mathbf{R}_j} \, \delta\mathbf{R}_j = \delta\mathbf{R}_j \, \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial\mathbf{R}_j\partial\mathbf{R}_i} \, \delta\mathbf{R}_i$$

the expansion can be written as

$$\begin{aligned} H_I(\{\mathbf{R}_l + \delta\mathbf{R}_l\}) = H_I(\{\mathbf{R}_l\}) &+ \sum_i \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial\mathbf{R}_i} \, \delta\mathbf{R}_i + \sum_{i,j;\, i\neq j} \delta\mathbf{R}_i \, \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial\mathbf{R}_i\partial\mathbf{R}_j} \, \delta\mathbf{R}_j \\ &+ \sum_{i,j;\, i\neq j} \delta\mathbf{R}_i \, \frac{\partial^2 V_{i,j}(\{\mathbf{R}_l\})}{\partial\mathbf{R}_i^2} \, \delta\mathbf{R}_i \end{aligned}$$

If I introduce a matrix U the expanded Hamiltonian can be brought into the same form as in the paper.

$$H_I(\{\mathbf{R}_l + \delta\mathbf{R}_l\}) = H_I(\{\mathbf{R}_l\}) + \sum_n \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial\mathbf{R}_n} \, \delta\mathbf{R}_n + \sum_{n,m} \delta\mathbf{R}_n \, U_{n,m} \, \delta\mathbf{R}_m$$

with

$$\begin{aligned} U_{n,m} &= \frac{\partial^2 V_{n,m}(\{\mathbf{R}_l\})}{\partial\mathbf{R}_n\partial\mathbf{R}_m} \quad \text{if } n \neq m \\ U_{n,n} &= \sum_{j;\, j\neq n} \frac{\partial^2 V_{n,j}(\{\mathbf{R}_l\})}{\partial\mathbf{R}_n^2} \end{aligned}$$

Swithing again to the reduced writting, where V_{ij} is only writen as a function of the variables of which it actually depends, reads

$$\begin{aligned} U_{n,m} &= \frac{\partial^2 V_{n,m}(\mathbf{R}_n, \mathbf{R}_m)}{\partial\mathbf{R}_n\partial\mathbf{R}_m} \quad \text{if } n \neq m \\ U_{n,n} &= \sum_{j;\, j\neq n} \frac{\partial^2 V_{n,j}(\mathbf{R}_n, \mathbf{R}_j)}{\partial\mathbf{R}_n^2} \end{aligned}$$

This result is a good point to further specify the potential that is used in the next sections. The potential will only depend on the relative vector between the two positions, i.e. $\mathbf{r}_{ij} = \mathbf{R}_i - \mathbf{R}_j$. Using the derivative

$$\frac{\partial\mathbf{r}_{ij}}{\partial\mathbf{R}_i} = \mathbb{1} = -\frac{\partial\mathbf{r}_{ij}}{\partial\mathbf{R}_j}$$

the derivatives of the U matrix can be written as

$$\begin{aligned} \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial\mathbf{R}_n\partial\mathbf{R}_m} &= \frac{\partial}{\partial\mathbf{R}_n} \left(\frac{\partial V_{n,m}(\mathbf{r}_{nm})}{\partial\mathbf{r}_{nm}} \, \frac{\partial\mathbf{r}_{nm}}{\partial\mathbf{R}_m} \right) \\ &= \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial\mathbf{r}_{nm}^2} \, \frac{\partial\mathbf{r}_{nm}}{\partial\mathbf{R}_m} \, \frac{\partial\mathbf{r}_{nm}}{\partial\mathbf{R}_n} + \frac{\partial V_{n,m}(\mathbf{r}_{nm})}{\partial\mathbf{r}_{nm}} \, \frac{\partial^2\mathbf{r}_{nm}}{\partial\mathbf{R}_n\partial\mathbf{R}_m} \\ &= -\frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial\mathbf{r}_{nm}^2} \end{aligned}$$

and

$$\begin{aligned} \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial\mathbf{R}_m^2 m} &= \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial\mathbf{r}_{nm}^2} \, \frac{\partial\mathbf{r}_{nm}}{\partial\mathbf{R}_m} \, \frac{\partial\mathbf{r}_{nm}}{\partial\mathbf{R}_m} \\ &= \frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial\mathbf{r}_{nm}^2} \end{aligned}$$

Thus

$$\begin{aligned} U_{n,m} &= -\frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial\mathbf{r}_{nm}^2} \quad \text{if } n \neq m \\ U_{n,n} &= \sum_{j;\, j\neq n} \frac{\partial^2 V_{n,j}(\mathbf{r}_{nj})}{\partial\mathbf{r}_{nj}^2} \end{aligned}$$

At this point I can eliminate the linear term of the Hamiltonian by introducing the trap potential

$$H_0(\{\delta\mathbf{R}_l\}) = \sum_n \frac{1}{2} m\omega^2\, \delta\mathbf{R}_n^2$$

and redefining the trap centers by a equal shift of all positions $\{\delta\mathbf{R}_l\} = \{\delta\tilde{\mathbf{R}}_l + \mathbf{\Delta}\}$. Neglection all constant terms the Hamiltonian up to second order reads

$$\begin{aligned} H(\{\delta\mathbf{R}_l\}) &= \sum_n \frac{1}{2} m\omega^2\, (\delta\tilde{\mathbf{R}}_n + \mathbf{\Delta}_n)^2 + \sum_n \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n}\, (\delta\tilde{\mathbf{R}}_n + \mathbf{\Delta}_n) + \sum_{n,m} (\delta\tilde{\mathbf{R}}_n + \mathbf{\Delta}_n)\, U_{n,m}\, (\delta\tilde{\mathbf{R}}_m + \mathbf{\Delta}_m) \\ &= \sum_n \frac{1}{2} m\omega^2\, \delta\tilde{\mathbf{R}}_n^2 + \sum_{n,m} \delta\tilde{\mathbf{R}}_n\, U_{n,m}\, \delta\tilde{\mathbf{R}}_m + \sum_n \left(\frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n} + \left(2 \sum_m \delta\tilde{\mathbf{R}}_m\, U_{m,n} + m\omega^2 \right) \mathbf{\Delta}_n \right) \delta\tilde{\mathbf{R}}_n, \end{aligned}$$

where it was used that U is symmetric. The linear term vanishes, if $\mathbf{\Delta}_n$ is chosen such that

$$\mathbf{\Delta}_n = -\left(2 \sum_m \delta\tilde{\mathbf{R}}_m\, U_{m,n} + m\omega^2 \right)^{-1} \frac{\partial H_I(\{\mathbf{R}_l\})}{\partial \mathbf{R}_n}$$

Thus

$$H(\{\delta\tilde{\mathbf{R}}_l\}) = \sum_n \frac{1}{2} m\omega^2\, \delta\tilde{\mathbf{R}}_n^2 + \sum_{n,m} \delta\tilde{\mathbf{R}}_n\, U_{n,m}\, \delta\tilde{\mathbf{R}}_m$$

To describe the deviation from the center in a quantum mechanical way, the displacements are now quantized as phonons using the quadrature operators

$$\delta\tilde{\mathbf{R}}_n = \frac{1}{\sqrt{m\omega}}\, (a_n + a_n^\dagger)$$

If we exploit that the interactions realized in U are much smaller compared to the trap frequency, i.e. $U_{n,m} \ll m\omega^2$, the creation and annhiliation of phonons is supressed and the dynamics only involves phonon hopping, thus combinations aa or $a^\dagger a^\dagger$ are discarded.

$$\begin{aligned} H &= \sum_n \omega\, a_n^\dagger a_n + 2 \sum_{n,m} \frac{U_{n,m}}{m\omega}\, a_n^\dagger a_m \\ &=: \sum_n \omega\, a_n^\dagger a_n + \sum_{n,m} \tilde{U}_{n,m}\, a_n^\dagger a_m \end{aligned}$$

Momentum space Hamiltonian

Let $c^A_{\mathbf{R},\alpha}$ and $c^B_{\mathbf{R},\alpha}$ be the phonon annihilation operators of sublattice A and B respectively at the unit cell, located at $\mathbf{R}$, where $\alpha \in \{x, y\}$ labels the the direction of movement. Let further L be the set of all unit cells of the lattice, and $C = \{c^A_{\mathbf{R},\alpha}, c^B_{\mathbf{R},\alpha}\}_{\mathbf{R}\in L, \alpha\in\{x,y\}}$ the set of all annihilation operators. The Hamiltonian in real space can then be written as

$$H = \sum_{\substack{\mathbf{R},\mathbf{R}'\in L \\ a,b\in C}} H^{a,b}(\mathbf{R} - \mathbf{R}')\, a^\dagger_{\mathbf{R}} b_{\mathbf{R}'}$$

For two dimensional periodic boundary conditions one can introduce the Fourier transformed operators, $a \in C$

$$a_{\mathbf{k}} = \frac{1}{N} \sum_{\mathbf{R}\in L} e^{-i\mathbf{k}\cdot\mathbf{R}}\, a_{\mathbf{R}},$$

$k \in \bar{L}$ where $\bar{L}$ is the reciprocal lattice of L and N is the extent of the lattice in one direction. Writing the Hamiltonian in terms of these operators yields

$$
\begin{aligned}
H &= \frac{1}{N^2} \sum_{\substack{\mathbf{R},\mathbf{R}' \in L \\ a,b \in C \\ \mathbf{k},\mathbf{k}' \in \bar{L}}} H^{a,b}(\mathbf{R}-\mathbf{R}')\, e^{-i\,\mathbf{k}\mathbf{R}}\, e^{i\,\mathbf{k}'\mathbf{R}'}\, a^\dagger_{\mathbf{k}} b_{\mathbf{k}'} \\
&= \frac{1}{N^2} \sum_{\substack{\mathbf{R},\ \Delta\mathbf{R}=\mathbf{R}-\mathbf{R}' \\ a,b \\ \mathbf{k},\mathbf{k}'}} H^{a,b}(\Delta\mathbf{R})\, e^{-i\,\mathbf{R}(\mathbf{k}-\mathbf{k}')}\, e^{-i\,\Delta\mathbf{R}\mathbf{k}'}\, a^\dagger_{\mathbf{k}} b_{\mathbf{k}'} \\
&= \sum_{\substack{\Delta\mathbf{R},\mathbf{k} \\ a,b}} H^{a,b}(\Delta\mathbf{R})\, e^{-i\,\Delta\mathbf{R}\mathbf{k}}\, a^\dagger_{\mathbf{k}} b_{\mathbf{k}} =: \sum_{\substack{\mathbf{k} \\ a,b}} H^{a,b}_{\mathbf{k}}\, a^\dagger_{\mathbf{k}} b_{\mathbf{k}} \\
\text{where} \quad H^{a,b}_{\mathbf{k}} &= \sum_{\Delta\mathbf{R}} H^{a,b}(\Delta\mathbf{R})\, e^{-i\,\Delta\mathbf{R}\mathbf{k}}
\end{aligned}
$$

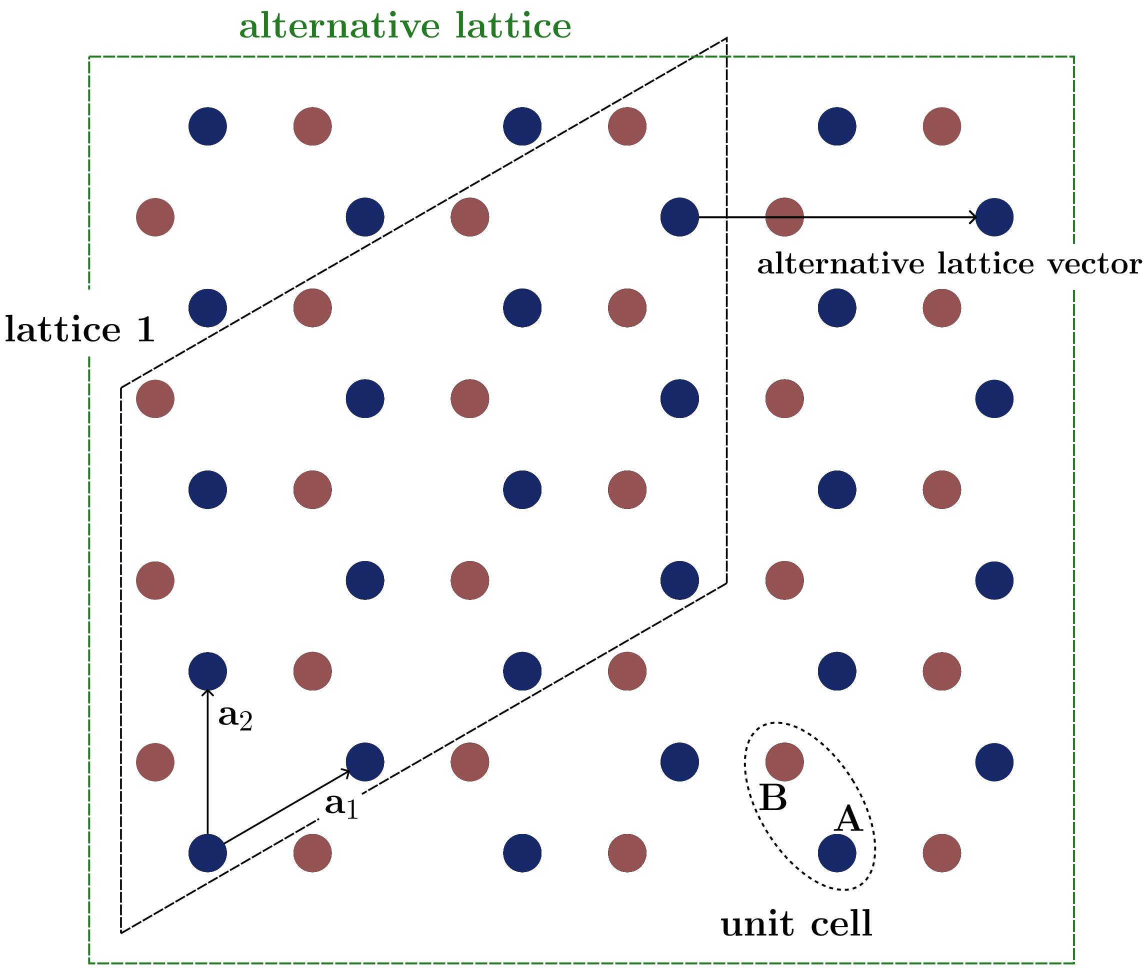

Figure 1: **Lattice definition** The main configuration of the grid,lattice 1, is the parallelogram spanned by the lattice vectors $\mathbf{a}_1$ and $\mathbf{a}_2$. This is visualized by the black dashed frame. There is also the possibility of defining a square lattice which is highlighted by the green dashed frame.

For the case where only one direction of the lattice obeys periodic boundary conditions, while the other direction has open boundaries, the Fourier transform is only performed in the periodic direction. For this purpose I introduce the lattice vectors $\mathbf{a}_1$ and $\mathbf{a}_2$, which span the lattice. The lattice can be split into subspaces $L_n = \{j \cdot \mathbf{a}_n\}_{j \in M_n}$ whith $n \in \{1,2\}$ and M_n is chosen in such a way, that L_n spans from one edge of the lattice to the other. Hence the position of any unit cell $\mathbf{R}$ can be expressed as the linear combination of two vectors $\mathbf{r}_1 \in L_1$ and $\mathbf{r}_2 \in L_2$, i.e. $\mathbf{R} = \mathbf{r}_1 + \mathbf{r}_2$. I can now also introduce the reciprocal lattice vectors $\mathbf{k}_1$ and $\mathbf{k}_2$ with the property $\mathbf{a}_i \cdot \mathbf{k}_j = 2\pi\delta_{ij}$. I perform the Fourier transformation as an example for the lattice 1 from Fig. 1.

Here the lattice direction represented by $\mathbf{a}_2$ shall be periodic. I introduce the one dimensionally Fourier transformed operators

$$a_{\mathbf{r}_1,\mathbf{k}} = \frac{1}{\sqrt{N}} \sum_{\mathbf{r}_2 \in L_2} e^{-i\mathbf{k}\mathbf{r}_2}\, a_{\mathbf{r}_1+\mathbf{r}_2}, \quad \mathbf{k} \in \{\frac{1}{N}, \dots, 1\} \cdot \mathbf{k}_2$$

The Hamiltonian can now be transformed via

$$\begin{aligned}
H &= \frac{1}{N} \sum_{\substack{\mathbf{r}_1,\mathbf{r}_1' \in L_1 \\ \mathbf{r}_2,\mathbf{r}_2' \in L_2}} \sum_{\substack{a,b \in C \\ \mathbf{k},\mathbf{k}' \in \bar{L}}} H^{a,b}(\mathbf{R}-\mathbf{R}')\, e^{-i\,\mathbf{k}\mathbf{r}_2}\, e^{i\,\mathbf{k}'\mathbf{r}_2'}\, a^\dagger_{\mathbf{r}_1,\mathbf{k}} b_{\mathbf{r}_1',\mathbf{k}'} \\
&= \frac{1}{N} \sum_{\substack{\mathbf{r}_1,\mathbf{r}_1' \\ \mathbf{r}_2,\ \Delta\mathbf{r}_2 = \mathbf{r}_2 - \mathbf{r}_2'}} \sum_{\substack{a,b \\ \mathbf{k},\mathbf{k}'}} H^{a,b}(\Delta\mathbf{R})\, e^{-i\,\mathbf{R}(\mathbf{k}-\mathbf{k}')}\, e^{-i\,\Delta\mathbf{R}\mathbf{k}'}\, a^\dagger_{\mathbf{r}_1,\mathbf{k}} b_{\mathbf{r}_1',\mathbf{k}'} \\
&= \sum_{\substack{\mathbf{r}_1,\mathbf{r}_1' \\ \Delta\mathbf{r}_2,\mathbf{k} \\ a,b}} H^{a,b}(\Delta\mathbf{R})\, e^{-i\,\Delta\mathbf{r}_2\mathbf{k}}\, a^\dagger_{\mathbf{r}_1,\mathbf{k}} b_{\mathbf{r}_1',\mathbf{k}} =: \sum_{\substack{\mathbf{r}_1,\mathbf{r}_1' \\ \mathbf{k},a,b}} H^{a,b}(\mathbf{k},\Delta\mathbf{R})\, a^\dagger_{\mathbf{r}_1,\mathbf{k}} b_{\mathbf{r}_1',\mathbf{k}}
\end{aligned}$$

$$\text{where} \quad H^{a,b}(\mathbf{k},\Delta\mathbf{R}) = \sum_{\Delta\mathbf{r}_2} H^{a,b}(\Delta\mathbf{R})\, e^{-i\,\Delta\mathbf{r}_2\mathbf{k}} = \sum_{\Delta\mathbf{r}_2} H^{a,b}(\Delta\mathbf{R})\, e^{-i\,\Delta\mathbf{R}\mathbf{k}}$$

The last equality holds, when $\mathbf{k}$ is in the direction of one of the reciprocal lattice vectors. However, as we perform only a one dimensional Fourier transform, I think a restriction to a purely one dimensional treatment is necessary. Thus $\mathbf{k}_n$ should only be chosen with respect to the condition $\mathbf{k}_n\mathbf{a}_n = 2\pi$, which gives the natural choice $\mathbf{k}_n \parallel \mathbf{a}_n$. For the case of the alternative square lattice I think, we should consider the number of rows twice as big as the number of columns and choose an alternative lattice vector into the x-direction, as depicted in Fig. 1.

What the one dimensional Fourier transform does, is projecting the system onto the one-dimensional chain not of atoms, but of interacting rows or columns of atoms. In this sense the division into sub-directions or sub-bases does not have to be restricted to the lattice vectors. The only thing needed is a labeling of the different rows or columns with the complementary direction having a periodic structure. This allows the division of the alternative square lattice as described above.

Aligned Dipole-Dipole Interactions

In this section the derivative of the dipole interaction will be performed in the case where all dipoles point into the same direction. The subsripts lableling the different positions are droped in order to make the notation plainer.

$$V_{dd}(\mathbf{r}) = V_0 \left(\frac{1}{r^3} - 3\,\frac{(\mathbf{r}\cdot\hat{\mathbf{d}})^2}{r^5} \right),$$

with $\mathbf{r} = \mathbf{R}_i - \mathbf{R}_j$ beeing the distance between any to sites i and j $(i \neq j)$. The first derivative with respect to component α of $\mathbf{r}$, $\alpha \in \{x, y, z\}$ reads

$$\frac{1}{V_0}\frac{\partial V_{dd}}{\partial r_\alpha} = -\frac{3r_\alpha}{r^5} + \frac{15 r_\alpha (\mathbf{r}\cdot\hat{\mathbf{d}})^2}{r^7} - \frac{6(\mathbf{r}\cdot\hat{\mathbf{d}})d_\alpha}{r^5},$$

the second derivative reads consequently

$$
\begin{aligned}
\frac{1}{V_0}\,\frac{\partial^2 V_{dd}}{\partial r_\alpha \partial r_\beta} &= -3\,\frac{\delta_{\alpha\beta}}{r^5} + 15\,\frac{r_\alpha r_\beta}{r^7} + 15\,\frac{\delta_{\alpha\beta}(\mathbf{r}\cdot\hat{\mathbf{d}})^2}{r^7} - 15\cdot 7\,\frac{r_\alpha r_\beta(\mathbf{r}\cdot\hat{\mathbf{d}})^2}{r^9} \\
&+ 30\,\frac{d_\alpha r_\beta + d_\beta r_\alpha}{r^7}\,(\mathbf{r}\cdot\hat{\mathbf{d}}) - 6\,\frac{d_\alpha d_\beta}{r^5} \\
\\
&= \delta_{\alpha\beta}\left(-3\,\frac{1}{r^5} + 15\,\frac{(\mathbf{r}\cdot\hat{\mathbf{d}})^2}{r^7}\right) - 6\,\frac{d_\alpha d_\beta}{r^5} + 30\,\frac{d_\alpha r_\beta + d_\beta r_\alpha}{r^7}\,(\mathbf{r}\cdot\hat{\mathbf{d}}) \\
&+ 15\,r_\alpha r_\beta\left(\frac{1}{r^7} - 7\,\frac{(\mathbf{r}\cdot\hat{\mathbf{d}})^2}{r^9}\right).
\end{aligned}
$$

General Dipole-Dipole Interactions

In the case where not all dipole moments are aligned, the properties of the interaction get extended to

$$
V_{dd}(\mathbf{r}) = V_0\,\left(\frac{\hat{\mathbf{d}}_1\cdot\hat{\mathbf{d}}_2}{r^3} - 3\,\frac{(\mathbf{r}\cdot\hat{\mathbf{d}}_1)(\mathbf{r}\cdot\hat{\mathbf{d}}_2)}{r^5}\right).
$$

The first derivative reads

$$
\frac{1}{V_0}\,\frac{\partial V_{dd}}{\partial r_\alpha} = -\frac{3\,(\hat{\mathbf{d}}_1\cdot\hat{\mathbf{d}}_2)\,r_\alpha}{r^5} + \frac{15 r_\alpha(\mathbf{r}\cdot\hat{\mathbf{d}}_1)(\mathbf{r}\cdot\hat{\mathbf{d}}_2)}{r^7} - \frac{3\,\left((\mathbf{r}\cdot\hat{\mathbf{d}}_1)\,d_{2,\alpha} + (\mathbf{r}\cdot\hat{\mathbf{d}}_2)\,d_{1,\alpha}\right)}{r^5},
$$

the second derivative reads consequently

$$
\begin{aligned}
\frac{1}{V_0}\,\frac{\partial^2 V_{dd}}{\partial r_\alpha \partial r_\beta} &= -3\,\frac{\delta_{\alpha\beta}\,(\hat{\mathbf{d}}_1\cdot\hat{\mathbf{d}}_2)}{r^5} + 15\,\frac{r_\alpha r_\beta\,(\hat{\mathbf{d}}_1\cdot\hat{\mathbf{d}}_2)}{r^7} + 15\,\frac{\delta_{\alpha\beta}(\mathbf{r}\cdot\hat{\mathbf{d}}_1)(\mathbf{r}\cdot\hat{\mathbf{d}}_2)}{r^7} - 15\cdot 7\,\frac{r_\alpha r_\beta(\mathbf{r}\cdot\hat{\mathbf{d}}_1)(\mathbf{r}\cdot\hat{\mathbf{d}}_2)}{r^9} \\
&+ 15\,\left[\frac{d_{1,\alpha} r_\beta + d_{1,\beta} r_\alpha}{r^7}\,(\mathbf{r}\cdot\hat{\mathbf{d}}_2) + \frac{d_{2,\alpha} r_\beta + d_{2,\beta} r_\alpha}{r^7}\,(\mathbf{r}\cdot\hat{\mathbf{d}}_1)\right] - 3\,\frac{d_{1,\alpha} d_{2,\beta} + d_{2,\alpha} d_{1,\beta}}{r^5} \\
\\
&= \delta_{\alpha\beta}\left(-3\,\frac{\hat{\mathbf{d}}_1\cdot\hat{\mathbf{d}}_2}{r^5} + 15\,\frac{(\mathbf{r}\cdot\hat{\mathbf{d}}_1)(\mathbf{r}\cdot\hat{\mathbf{d}}_2)}{r^7}\right) - 3\,\frac{d_{1,\alpha} d_{2,\beta} + d_{2,\alpha} d_{1,\beta}}{r^5} \\
&+ 15\,\left[\frac{d_{1,\alpha} r_\beta + d_{1,\beta} r_\alpha}{r^7}\,(\mathbf{r}\cdot\hat{\mathbf{d}}_2) + \frac{d_{2,\alpha} r_\beta + d_{2,\beta} r_\alpha}{r^7}\,(\mathbf{r}\cdot\hat{\mathbf{d}}_1)\right] \\
&+ 15\,r_\alpha r_\beta\left(\frac{\hat{\mathbf{d}}_1\cdot\hat{\mathbf{d}}_2}{r^7} - 7\,\frac{(\mathbf{r}\cdot\hat{\mathbf{d}}_1)(\mathbf{r}\cdot\hat{\mathbf{d}}_2)}{r^9}\right).
\end{aligned}
$$

For the the Van der Waals interaction the potential has to be squared. Therefore the second derivative reads

$$
\begin{aligned}
\frac{\partial V_{dd}^2}{\partial r_\alpha} &= 2\,V_{dd}\frac{\partial V_{dd}}{\partial r_\alpha} \\
\frac{\partial^2 V_{dd}^2}{\partial r_\alpha \partial r_\beta} &= 2\,\frac{\partial V_{dd}}{\partial r_\alpha}\frac{\partial V_{dd}}{\partial r_\beta} + 2\,V_{dd}\frac{\partial^2 V_{dd}}{\partial r_\alpha \partial r_\beta}
\end{aligned}
$$

Winding Number

A chiral Hamiltonian is defined through the property, that it anticommutes with a unitary hermitian operator Γ

$$
\Gamma H \Gamma = -H \quad \text{with} \quad \Gamma^2 = 1. \tag{0.1}
$$

From this property follows that the Hamiltonian can be written in block off-diagonal form

$$H = \begin{pmatrix} 0 & h \\ h^\dagger & 0 \end{pmatrix}, \tag{0.2}$$

where h is, in general, a non-Hermitian matrix. This structure also applies for the matrix in momentum space and allows the definition of a winding number, which characterizes the topological properties of the system. If the Hamiltonian is 2×2 dimensional, the winding number can be defined through the decomposition of H in terms of the Pauli matrices

$$H = \mathbf{n} \cdot \boldsymbol{\sigma}, \tag{0.3}$$

where $\mathbf{n} = (n_x, n_y, 0)$ is a vector in the xy-plane. The winding number is then given by the winding of the vector $\mathbf{n}$ around the origin when traversing the Brillouin zone. Identifying the xy-plane of $\mathbf{n}$ with the complex plane, which defines a path $\mathbf{n}(k) \to \gamma(k)$, the winding number can be calculated through

$$w = \frac{1}{2\pi i} \oint_{\gamma_k} \frac{1}{z}\, dz = \frac{1}{2\pi i} \int_{\mathrm{BZ}} dk \frac{1}{\gamma(k)} \frac{d\gamma(k)}{dk}.$$

In terms of the Hamiltonian above this reads

$$w = \frac{1}{2\pi i} \int_{\mathrm{BZ}} dk\, \frac{h'(k)}{h(k)}.$$

Generalizing the winding number to higher dimensional Hamiltonians is possible through the use of the block off-diagonal form. The winding number then reads [6, 7]

$$w = \frac{1}{2\pi i} \int_{\mathrm{BZ}} dk\, \mathrm{Tr}\left[h^{-1}(k)\partial_k h(k)\right].$$

Diagonalizing h as $h = S\Lambda S^{-1}$ with the diagonal matrix $\Lambda = \mathrm{diag}(\lambda_1, \lambda_2, ...)$ and the invertible matrix S, the trace can be rewritten as

$$\begin{aligned}
\mathrm{Tr}\left[h^{-1}\partial_k h\right] &= \mathrm{Tr}\left[S\Lambda^{-1}S^{-1}\partial_k(S\Lambda S^{-1})\right] \\
&= \mathrm{Tr}\left[\Lambda^{-1}\partial_k\Lambda + S^{-1}\partial_k S + (\partial_k S^{-1})S\right] \\
\mathrm{Tr}\left[S^{-1}\partial_k S + S\partial_k S^{-1}\right] &= \mathrm{Tr}\left[\partial_k\left(SS^{-1}\right)\right] = 0, \\
\Rightarrow \mathrm{Tr}\left[h^{-1}\partial_k h\right] &= \mathrm{Tr}\left[\Lambda^{-1}\partial_k\Lambda\right] = \sum_i \frac{\partial_k\lambda_i}{\lambda_i} = \partial_k\left(\sum_i \ln(\lambda_i)\right) = \partial_k \ln(\det[h]).
\end{aligned}$$

Another useful definition of the winding number is through the so-called Q-matrix, which is defined through the projectors onto the eigenstates of the Hamiltonian. As the Hamiltonian is chiral, the eigenvalues come in pairs $\pm E$. We introduce a labeling of the eigenvalues, such that $E_j = -E_{-j}$ with $j = 1, 2, ..., N$ and N being half the dimension of the Hamiltonian. The corresponding eigenstates are labeled accordingly ($|\psi_j\rangle$) and posses the property $\Gamma|\psi_j\rangle = |\psi_{-j}\rangle$. The Q-matrix is defined as

$$Q = \sum_{j=1}^{N}\left(|\psi_j\rangle\langle\psi_j| - |\psi_{-j}\rangle\langle\psi_{-j}|\right).$$

As a consequence of the chiral symmetry, the Q-matrix anticommutes with Γ and can therefore be written in block off-diagonal form in the basis, where Γ is diagonal, which is also the case in which H is block off-diagonal.

$$\begin{aligned}
\Gamma Q\Gamma &= \sum_{j=1}^{N}\left(|\psi_{-j}\rangle\langle\psi_{-j}| - |\psi_j\rangle\langle\psi_j|\right) = -Q \\
\Rightarrow Q &= \begin{pmatrix} 0 & q \\ q^\dagger & 0 \end{pmatrix}
\end{aligned}$$

The fact that the winding number of the q-matrix is the same as that of h can be shown with the help of the following two properties.

$$\begin{aligned} Q^2 = 1 \quad &\Rightarrow \quad qq^\dagger = \mathbb{1}, \quad q^\dagger q = \mathbb{1} \\ QHQ = H \quad &\Rightarrow \quad qh^\dagger q = h^\dagger \end{aligned}$$

From the second property follows that $q^\dagger h = h^\dagger q$. Thus $h^\dagger q$ is a hermitian matrix and therefore has no winding number, since the eigenvalues are real. Hence

$$\begin{aligned} 0 &= \mathrm{Tr}\left[q^\dagger (h^\dagger)^{-1}\partial_k (h^\dagger q)\right] \\ &= \mathrm{Tr}\left[q\, q^\dagger\, (h^\dagger)^{-1}\partial_k h^\dagger + h^\dagger (h^\dagger)^{-1}\, q^\dagger \partial_k q\right] \\ \Rightarrow \mathrm{Tr}\left[q^{-1}\partial_k q\right] &= -\mathrm{Tr}\left[(h^\dagger)^{-1}\partial_k h^\dagger\right] \\ &= \mathrm{Tr}\left[h^{-1}\partial_k h\right] \end{aligned}$$

There is another useful property, for which we consider the basis in which $\Gamma = \sigma_z \otimes \mathbb{1}_N$. This is the same diagonal basis as previously used. In this basis we can represent all eigenvalues of H as a stack of to N-component vectors. The eigenvector for negative energies is written as

$$|\psi_i^-\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} |h_{1i}\rangle \\ |h_{2i}\rangle \end{pmatrix} \qquad i \in \{1, \dots, N\}$$

Because of the chirality of the Hamiltonian the eigenvector with positive energy is already determined by this choice and reads

$$|\psi_i^+\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} |h_{1i}\rangle \\ -\,|h_{2i}\rangle \end{pmatrix}$$

The matrix Q is represented in the following form

$$\begin{aligned} Q &= \sum_i \begin{pmatrix} 0 & |h_{1i}\rangle \langle h_{2i}| \\ |h_{2i}\rangle \langle h_{1i}| & 0 \end{pmatrix} \\ \Rightarrow \quad q &= \sum_i |h_{1i}\rangle \langle h_{2i}| \end{aligned}$$

If we now compute the kernel of the winding number with respect to q we get

$$\begin{aligned} \mathrm{Tr}[q^\dagger \partial_k q] &= \mathrm{Tr}\left[\sum_{i,j} |h_{2i}\rangle \langle h_{1i}|\, \partial_k\, \left(|h_{1j}\rangle \langle h_{2j}|\right)\right] \\ &= \sum_{i,j} \mathrm{Tr}\left[|h_{2i}\rangle \langle h_{1i}| \left(\partial_k |h_{1j}\rangle\right) \langle h_{2j}|\right] + \sum_{i,j} \mathrm{Tr}\left[|h_{2i}\rangle \langle h_{1i}|\, |h_{1j}\rangle\, \left(\partial_k \langle h_{2j}|\right)\right] \\ &= \sum_i \langle h_{1i}| \left(\partial_k |h_{1i}\rangle\right) + \sum_i \left(\partial_k \langle h_{2i}|\right) |h_{2i}\rangle \end{aligned}$$

When computing the winding number the second term contributes with a minus sign compared to its complex conjugated self, i.e.

$$\int_{\mathrm{BZ}} dk\, \left(\partial_k \langle h_{2i}|\right) |h_{2i}\rangle = -\int_{\mathrm{BZ}} dk\, \langle h_{2i}| \left(\partial_k |h_{2i}\rangle\right)$$

Thus the winding number reads

$$\begin{aligned} w &= \frac{1}{2\pi i}\int_{\mathrm{BZ}} dk\, \mathrm{Tr}[q^\dagger \partial_k q] \\ &= \frac{1}{2\pi i}\sum_i \int_{\mathrm{BZ}} dk\, \left(\langle h_{1i}|\, \partial_k\, |h_{1i}\rangle - \langle h_{2i}|\, \partial_k\, |h_{2i}\rangle\right) \end{aligned}$$

This can be expressed with respect to the full eigenvectors of the Hamiltonian via

$$
\begin{aligned}
w &= \frac{1}{\pi i} \sum_i \int_{\mathrm{BZ}} dk \, \langle \Gamma \psi_i^- | \, \partial_k \, | \psi_i^- \rangle \\
&= \frac{1}{\pi i} \sum_i \int_{\mathrm{BZ}} dk \, \langle \psi_i^+ | \, \partial_k \, | \psi_i^- \rangle \\
&= \frac{1}{2\pi i} \sum_i \int_{\mathrm{BZ}} dk \, \left(\langle h_{1i} | \, \partial_k \, | h_{1i} \rangle - \langle h_{2i} | \, \partial_k \, | h_{2i} \rangle \right)
\end{aligned}
$$

The quantity $\sum_i \langle \Gamma \psi_i^- | \, \partial_k \, | \psi_i^- \rangle$ can be related to a polarisation [8].

1D analysis of topological states

In order to have a chiral Hamiltonian I will choose the orientation of the dipoles in a way, such that the chirality remains. This will be done for each lattice configuration separately, i.e. the dipoles depend on the lattice geometry.
In order to make the achievement of chirality more feasible I only consider nearest neighbour interactions. When talking of chirality the chirality of the matrix U, defined above, is in mind. Recall that

$$
\begin{aligned}
U_{n,m} &= -\frac{\partial^2 V_{n,m}(\mathbf{r}_{nm})}{\partial \mathbf{r}_{nm}^2} \quad \text{if } n \neq m \\
U_{n,n} &= \sum_{j;\, j \neq n} \frac{\partial^2 V_{n,j}(\mathbf{r}_{nj})}{\partial \mathbf{r}_{nj}^2}
\end{aligned}
$$

Chirality implies that all couplings between sites on the same sublattice vanish. The nearest neighbour restriction already assures that couplings between different sites of the same sublattice vanish. Left is thus only the onsite potential due to the taylor expansion of the dipole dipole potential. The aim is now to choose the orientaiton of the dipoles in such a way that the sum in $U_{n,n}$ is equal to zero.
Be aware that so far I didn't write explicitely the dependence of U from the direction of atom movement x, y. Hence $U_{n,m}$, $U_{n,n}$ are themselves two times two matrixes $(U_{n,m}^{\alpha,\beta})_{\alpha,\beta \in \{x,y\}}$ and the Hamilonian expressed explicitely with respect to the x, y-phonon modes reads

$$
H = \sum_{n,\alpha} \omega \, a_{n,\alpha}^\dagger a_{n,\alpha} + \frac{2}{m\omega} \sum_{n,m,\alpha,\beta} U_{n,m}^{\alpha,\beta} \, a_{n,\alpha}^\dagger a_{m,\beta}
$$

The terms with $a_{n,\alpha}^\dagger a_{n,\alpha}$ can thus be eliminiated by redefining the trap frequency.

$$
\omega_\alpha = \omega + \frac{2}{m\omega} \, U_{n,n}^{\alpha,\alpha}
$$

The terms coupling the two directions of phonon modes on the same side can not be internalized by another quantity. Here the degrees of freedom related to the dipole orientation are exploited to tune the $U_{n,n}^{x,y}$ values to zero, as mentioned above. Because of the nearest neighbour nature of the consideration only two distances play a role, the intracell distance $\mathbf{r}_I$ and the intercell distance $\mathbf{r}_O$.

$$
U_{n,n}^{x,y} = \frac{\partial^2 V(\mathbf{r}_I)}{\partial r_{I,x} \partial r_{I,y}} + \frac{\partial^2 V(\mathbf{r}_O)}{\partial r_{O,x} \partial r_{O,y}}
$$

The dipoles are then chosen in such a way that this sum vanishes. For open boundary conditions the values of $U_{n,n}^{\alpha,\beta}$ at the borders are different from the bulk values. In the case of $\alpha = \beta$ the difference can be compensated by changed trap frequencies at the boundary. For the xy coupling this is not possible and the interaction due to both neighboring atoms has to made zero individually such that x- and y-modes are completely decoupled from each other.

The topologically non-trivial phase actually breaks down, when those border-terms are not eliminated. This is confirmed by simulations. It might be, at first, counter intuitive that reducing the effect of an open border is decremental to the observation of edge states. It makes, however perfect sense, when remembering that those boundary terms in our case destroy the chiral symmetry. It seems to be necessary for the bulk-boundary condition to hold that not only the momentum space Hamiltonian with periodic boundary conditions has to satisfy chiral symmetry but also the real space Hamiltonian with open boundary conditions. Otherwise one could choose the Hamiltonian describing the border in a completely arbitrary manner and still observe edge states, what would be quite a strong statement.

What one observes is that, if the same direction edge coupling is kept in, no separated energies in the bulk gaps arise. At the other hand if only the boundary terms, coupling the two phonon directions, remain and $U^{\alpha,\alpha}_{n,n}$ is completely absorbed into the trap frequencies separated energies inside the bulk energy gaps can be observed. These separated energies, however, are not located at zero and converge for the atom number going to infinity to a non-zero value, what makes them non-topological.

Aligned dipoles in plane

I consider a geometry with two sublattices repeating itself in the x-axis and separeted by the distance h from each other into the y-axis. Different configurations of this geometry are realized by moving the sublattices against each other in the x-direction. I consider an even amount of lattice, with N atoms in each sublattice. The nth atom of each sublattice A and B takes the position.

$$\begin{aligned}\mathbf{r}^A_n &= n \cdot a\,\mathbf{e}_x \\ \mathbf{r}^B_n &= n \cdot a\,\mathbf{e}_x + b\,\mathbf{e}_x + h\,\mathbf{e}_y,\end{aligned}$$

where a is the lattice constant, and the parameters b and h can be chosen to realize different lattice configurations with the restriction that b and h have to be chosen in a way, such that the nearest neighbor restriction makes sense. For any b-value it is possible to find h and the dipole moment $\mathbf{d}$ in a way that fulfils the above conditions.

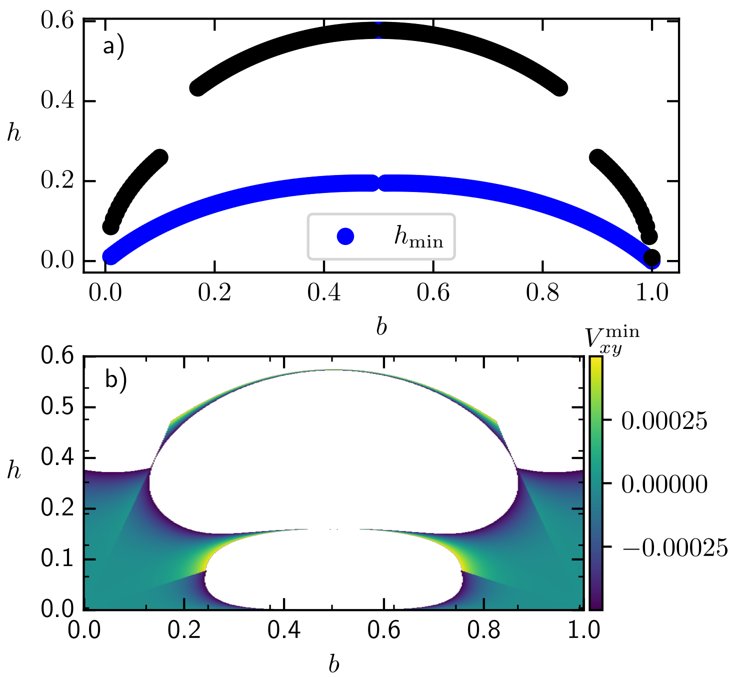

Figure 2: **Optimal h for chiral symmetry.** In **a)** The h values are depicted which decouple x- and y-phonons. Those values were determined numerically by computing roots of $V_xy(h)$. A grid of h values as initial guesses for the root algorithm was taken to have a better chance of finding all roots. In **b)** $V_xy(b, h)$ is drawn for reference. Larger values of the function have been omitted to make the pattern of small coupling values more visible.

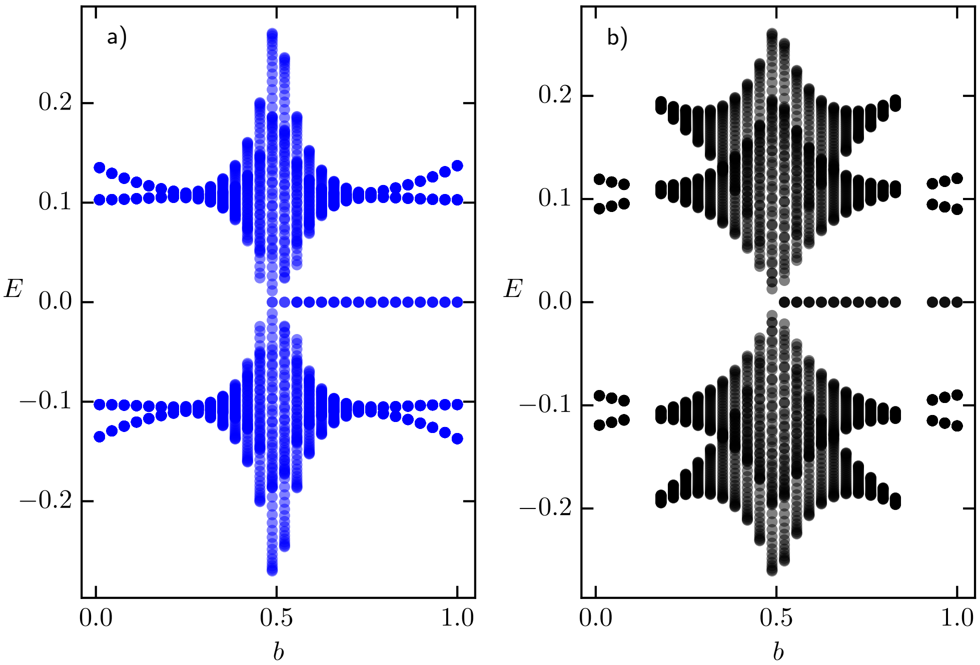

Figure 3: **Energy spectrum for the chiral Hamiltonian.**The diagram was created with the parameter $a = 1$. Blue and black colors depict the Energy spectrum for the h values picked from the blue and black curves in Fig. 2 respectively.

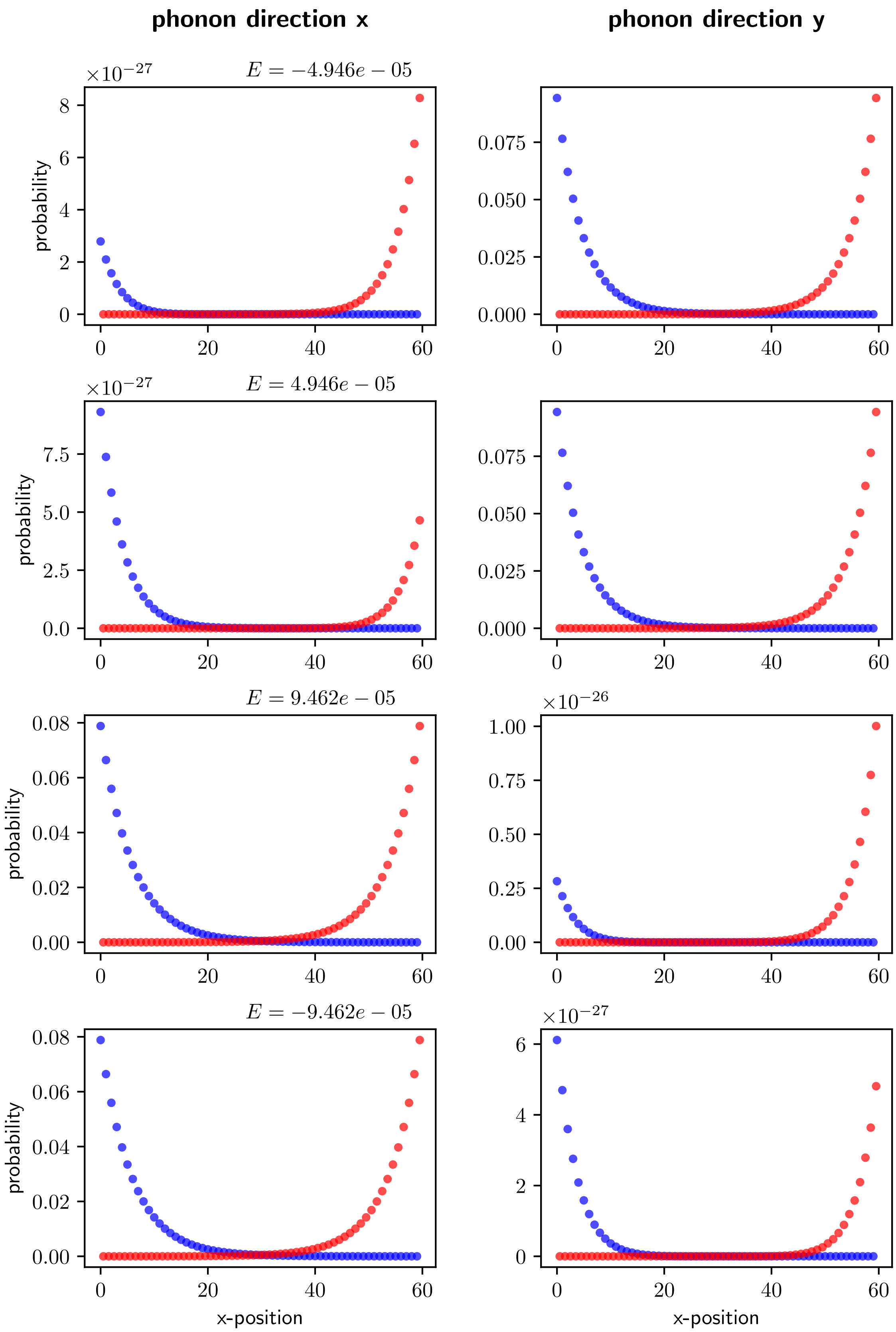

Figure 4: **Edge states**. The figure shows the decay of edge states with the four energies with the smallest absolute value. Each row depicts one energy state. Drawn in blue are the excitations of sublattice A, while the excitation of sublattice B is shown in red.

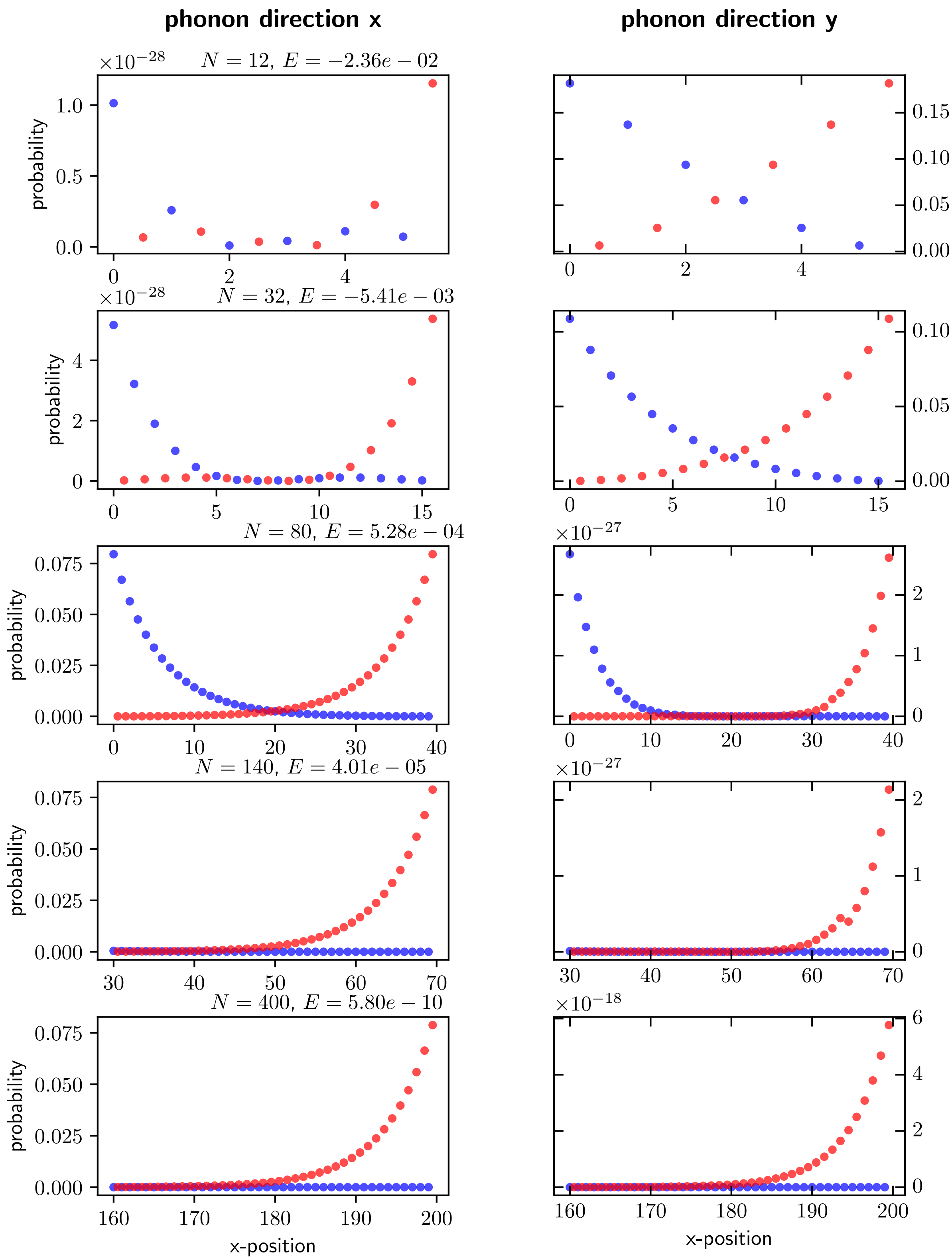

Figure 5: **Decay of energy for topological states with atom number.**. In each row the eigenstate for the energy with smallest absolute value is plotted. Blue and red show again the excitations of sublattice A and B respectively. When moving down in the rows of the figure the atom number of the system is increased. It shows that with increasing atom number the energy of the topological decays. What also can be observed. That the decay-length of the excitations when moving away from the edges does not change with particle number.

Spin-phonon coupling

Transport

I want to allow for the propagation of excitations through the chain. The first step towards this is constructing a Hamiltonian which describes the dynamics of a system of Rydberg atoms, where one atom is excited, hence restricting ourselves to the single excitation subspace. The general coupling between the atoms arises from the interaction between the dipole moments of the atoms,

$$V = \sum_{\mathbf{r}} V(\mathbf{r})$$
$$\text{with} \quad V(\mathbf{r}) = V_0 \left(\frac{\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2}{r^3} - 3\, \frac{(\mathbf{r} \cdot \hat{\mathbf{d}}_1)(\mathbf{r} \cdot \hat{\mathbf{d}}_2)}{r^5} \right).$$

Each individual atom is characterized by its position in the lattice and its atomic energy level which is captured by the non-interacting Hamiltonian H_0. The total Hamiltonian thus reads

$$H = H_0 + V$$

We will investigate the system in a regime, where the coupling due to the dipole forces is much smaller than the energy gap Δ, which separates the energy of the atoms from their next higher and lower states, i.e. $\|V\| \ll \Delta$. In order to gain an approximate effective Hamiltonian, which acts in a closed way on the one excitation subspace, I will perform a Schrieffer-Wolff transform up to second order [9]. Let P_0 be the projector on the one excitation subspace. Then an exact alternative Hamiltonian H' can be constructed by the transformation

$$H' = e^S\, H\, e^{-S}$$

for which holds, that it is block-diagonal with respect to the subspaces defined by the projections P_0 and $1 - P_0$. Thus the alternative Hamiltonian H' is closed with respect to the one excitation subspace and one can define the exact effective Hamiltonian as

$$H_{\text{eff}} = P_0\, e^S\, H\, e^{-S}\, P_0$$

Let ε be of the order of the coupling $\varepsilon = \|V\|$. The transformation generating operator S can be expanded in orders of V.

$$S = \sum_{n=1}^{\infty} \varepsilon^n\, S_n$$

I will consider the effective Hamiltonian up to second order in the coupling. This lets H_{eff} read

$$H_{\text{eff}} = H_0 P_0 + P_0 V P_0 + \frac{1}{2}\, P_0\, [S_1, V_{\text{od}}]\, P_0,$$

with $V_{\text{od}} = P_0\, V\, (1 - P_0) + (1 - P_0)\, V\, P_0$ being the off-diagonal part of the interaction. The first term of the generator S has the form

$$S_1 = \mathcal{L}(V_{\text{od}})$$
$$\text{with} \quad \mathcal{L}(X) = \sum_{\substack{i,j \in E \\ E_i \neq E_j}} \frac{\langle i|V|j\rangle}{E_i - E_j}\, |i\rangle\, \langle j|$$

The sum in the superoperator $\mathcal{L}$ goes over all energy eigenstates, where the i, j combination of states with equal energy are omitted. It can be written via a sum over all energies in the single excitations subspace $E1$ and all the other eigenenergies $\tilde{E}$.

$$\mathcal{L}(X) = \sum_{\substack{i\in E1\\ j\in\tilde{E}}} \frac{\langle i|V|j\rangle}{E_i - E_j} |i\rangle\langle j| - \text{h.c.}$$

From this the commutator term in the effective Hamiltonian can be calculated.

$$P_0 [S_1, V_{\text{od}}] P_0 = 2 \sum_{\substack{i\in E1\\ j\in\tilde{E}}} \frac{|\langle i|V|j\rangle|^2}{E_i - E_j} |i\rangle\langle i| =: H_{\text{eff},2}$$

Thus finally the effective Hamiltonian reads

$$H_{\text{eff}} = H_0 P_0 + P_0 V P_0 + \sum_{\substack{i\in E1\\ j\in\tilde{E}}} \frac{|\langle i|V|j\rangle|^2}{E_i - E_j} |i\rangle\langle i|$$

For the effective Hamiltonian the contribution $H_0 P_0$ is just a constant term and can be neglected. Further because of the large relative size of Δ over the interaction strength, I will only include energies into the above sum which differ from the single excitation subspace by Δ. Further the zero excitations subspace does not couple to the one with one excitation in the considered order of expansion. Thus only eigenstates in the double excitation subspace have to be taken into account for the sum.

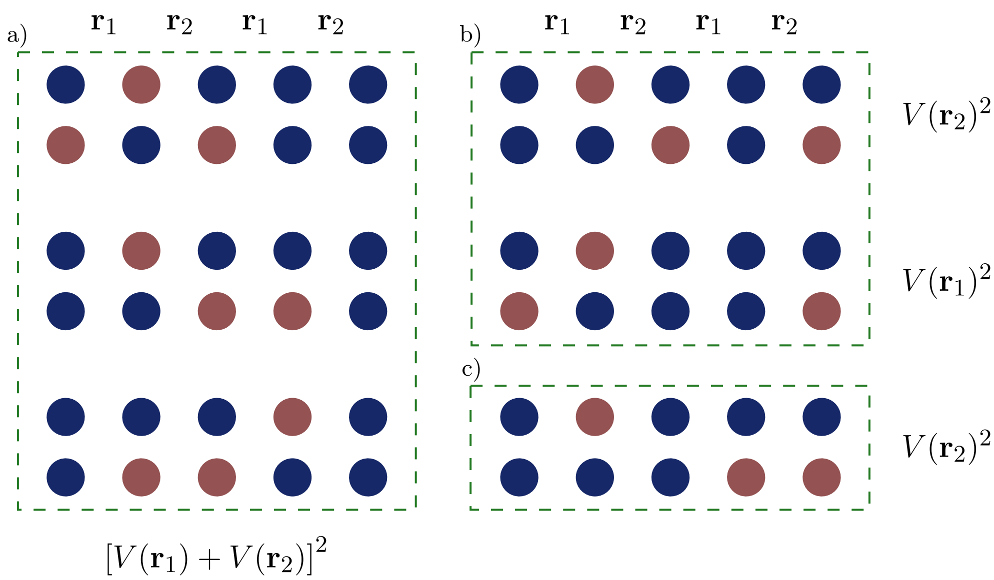

Figure 6: **Coupling schemes.** There are different ways double excited states couple to the single excited ones. The top row always depicts a state of the single, while the bottom row depicts a state of the double excited subspace. Generally configurations of two different subspaces are coupled to each other, when for two neighboring atoms one configuration has the inverted excitation scheme with respect to the other. I assign them to three categories. The first category is depicted in a). It couples two neighboring atoms twice. In the picture all possible cases are shown. For b) only two atoms couple two each other. Here the coupling takes place at a pair of $|e\rangle$ and $|g\rangle$ in both configurations. If the excitation in the single excitation subspace is at position $2 < i < N-1$, there are $N-4$ different states in the double excited subspace that contribute a term $V(\mathbf{r}_1)^2$ and the same number for $V(\mathbf{r}_2)^2$. Category c) couples two unexcited sites in the single excitations subspace to two excited sites in the double excited subspace. For this case there are $N/2-2$ states that contribute $V(\mathbf{r}_1)^2$ and $N/2-2$ states that contribute $V(\mathbf{r}_2)^2$, if the single excited state is is excited at atom $2 < i < N-1$.

In the following the effective Hamiltonian will be calculated for the specific configuration we are looking at. We consider a one-dimensional staggered chain of N atoms, where the atoms are separated by alternating distances $\mathbf{r}_1$ and $\mathbf{r}_2$. For a single atom there are only two states available, excited $|e\rangle$ and unexcited $|g\rangle$. The states $|i\rangle$, which are summed over, represent the states where only atom i is excited and the others are in the unexcited state. Going to a regime where only nearest neighbor interactions contribute we can write using the coupling schemes of Fig. 6

$$H_{\text{eff},2} = \sum_{i=1}^{N} \tilde{V}_i \; |i\rangle \langle i|$$

$$\tilde{V}_i = -\frac{1}{\Delta} \begin{cases} 3\;[V(\mathbf{r}_1)+V(\mathbf{r}_2)]^2 + (N-4+\frac{N}{2}-2)\,V^2(\mathbf{r}_1) + (N-4+\frac{N}{2}-2)\,V^2(\mathbf{r}_2) & \text{if } 1 \neq i \neq N \\ [V(\mathbf{r}_1)+V(\mathbf{r}_2)]^2 + (N-3+\frac{N}{2}-1)\,V^2(\mathbf{r}_1) + (\frac{N}{2}-1)\,V^2(\mathbf{r}_2) & \text{if } i=1 \\ 2\;[V(\mathbf{r}_1)+V(\mathbf{r}_2)]^2 + (N-3+\frac{N}{2}-2)\,V^2(\mathbf{r}_1) + (N-4+\frac{N}{2}-1)\,V^2(\mathbf{r}_2) & \text{if } i=2 \\ 2\;[V(\mathbf{r}_1)+V(\mathbf{r}_2)]^2 + (N-4+\frac{N}{2}-1)\,V^2(\mathbf{r}_1) + (N-3+\frac{N}{2}-2)\,V^2(\mathbf{r}_2) & \text{if } i=N-1 \\ [V(\mathbf{r}_1)+V(\mathbf{r}_2)]^2 + (\frac{N}{2}-1)\,V^2(\mathbf{r}_1) + (N-3+\frac{N}{2}-1)\,V^2(\mathbf{r}_2) & \text{if } i=N \end{cases}$$

The part involving interactions within the same subspace can be written as

$$P_0 V P_0 = \sum_{\substack{i=1\\ j=i+1}}^{N-1} V(\mathbf{r}_{i,j})\,\sigma_i^+\sigma_j^- + \text{h.c.},$$

where $\mathbf{r}_{i,j}$ is the distance between the ith and jth site. I now also want to write $H_{\text{eff},2}$ in terms of a sum of two atom operators. Therefore I introduce the projectors $n_i^e = |e\rangle_i \langle e|_i$ and $n_i^g = |g\rangle_i \langle g|_i$.

$$\begin{aligned} H_{\text{eff},2} = &\sum_{\substack{i=1\\ j=i+1}}^{N-1} V^2(\mathbf{r}_{i,j})\,(n_i^e n_j^g + n_i^g n_j^e + n_i^g n_j^g) + \sum_{i=2}^{N-1} [V(\mathbf{r}_{i-1,i}) + V(\mathbf{r}_{i,i+1})]^2 \; n_{i-1}^g n_i^e n_{i+1}^g \\ &+ \sum_{i=3}^{N-1} [V(\mathbf{r}_{i-2,i-1}) + V(\mathbf{r}_{i-1,i})]^2 \; n_{i-2}^g n_{i-1}^g n_i^e + \sum_{i=2}^{N-2} [V(\mathbf{r}_{i,i+1}) + V(\mathbf{r}_{i+1,i+2})]^2 \; n_i^e n_{i+1}^g n_{i+2}^g \\ &+ [V(\mathbf{r}_{1,2}) + V(\mathbf{r}_{2,3})]^2\, n_1^e n_2^g n_3^g + [V(\mathbf{r}_{N-2,N-1}) + V(\mathbf{r}_{N-1,N})]^2\, n_{N-2}^g n_{N-1}^g n_N^e \end{aligned}$$

A last step is possible since in the single excitation subspace terms like $n_i^e n_j^g$ simplify to just n_i^e.

$$\begin{aligned} &= \sum_{i=2}^{N-1} \Big(V^2(\mathbf{r}_{i-1,i}) + V^2(\mathbf{r}_{i,i+1}) + [V(\mathbf{r}_{i-1,i}) + V(\mathbf{r}_{i,i+1})]^2 \Big) \; n_i^e \\ &+ \sum_{i=3}^{N-1} [V(\mathbf{r}_{i-2,i-1}) + V(\mathbf{r}_{i-1,i})]^2 \; n_i^e + \sum_{i=2}^{N-2} [V(\mathbf{r}_{i,i+1}) + V(\mathbf{r}_{i+1,i+2})]^2 \; n_i^e \\ &+ \Big(V^2(\mathbf{r}_{1,2}) + [V(\mathbf{r}_{1,2}) + V(\mathbf{r}_{2,3})]^2 \Big) \; n_1^e + \Big(V^2(\mathbf{r}_{N-1,N}) + [V(\mathbf{r}_{N-2,N-1}) + V(\mathbf{r}_{N-1,N})]^2 \Big) \; n_N^e \\ &+ \sum_{i=1}^{N-1} V(\mathbf{r}_{i,i+1})\, n_i^g n_{i+1}^g \end{aligned}$$

First order approximation

For the first approach I will continue with the Hamiltonian up to first order in the dipole interaction, i.e. without $H_{\text{eff},2}$. The phonons are introduced by again expanding the interaction up to second order. I will restrict the consideration also to the constant phonon excitation subspace. For this reason the first order expansion of the dipole interaction will be ignored, as it couples subspaces with different phonon numbers. The Hamiltonian thus reads

$$H = \sum_{i=1}^{N-1} \left[V(\mathbf{r}_{i,i+1}) + \sum_{n,m\in\{i,i+1\}} \frac{\partial^2 V(\mathbf{r}_{i,i+1})}{\partial \mathbf{r}_n \partial \mathbf{r}_m}\, \delta\hat{\mathbf{r}}_n \delta\hat{\mathbf{r}}_m \right] \; \sigma_i^+ \sigma_{i+1}^- + \text{h.c.}$$

The hats in $\delta\hat{\mathbf{r}}$ indicates operators. Here also the operators are projected on the constant phonon subspace. What has to be accounted for is that this second order expansion is only valid if the displacement is small, which means $1/\sqrt{m\omega} \ll 1$ as $||a|| \sim 1$.

$$\left|1 - \frac{V(\mathbf{r} + \delta\mathbf{r})}{V(\mathbf{r})}\right| \ll 1$$

$$\left|\frac{\partial^2 V(\mathbf{r})}{\partial \mathbf{r}^2}\,\delta\mathbf{r}^2\right| \ll |V(\mathbf{r})|$$

When quantizing the displacements to phonon operators $\delta\mathbf{r}^2 \to \frac{2}{m\omega} a^\dagger a$, one finds the condition

$$\frac{2}{m\omega}\left|\frac{\partial^2 V(\mathbf{r})}{\partial \mathbf{r}^2}\right| \ll |V(\mathbf{r})|\,,$$

as $a^\dagger a$ is of the order of 1.

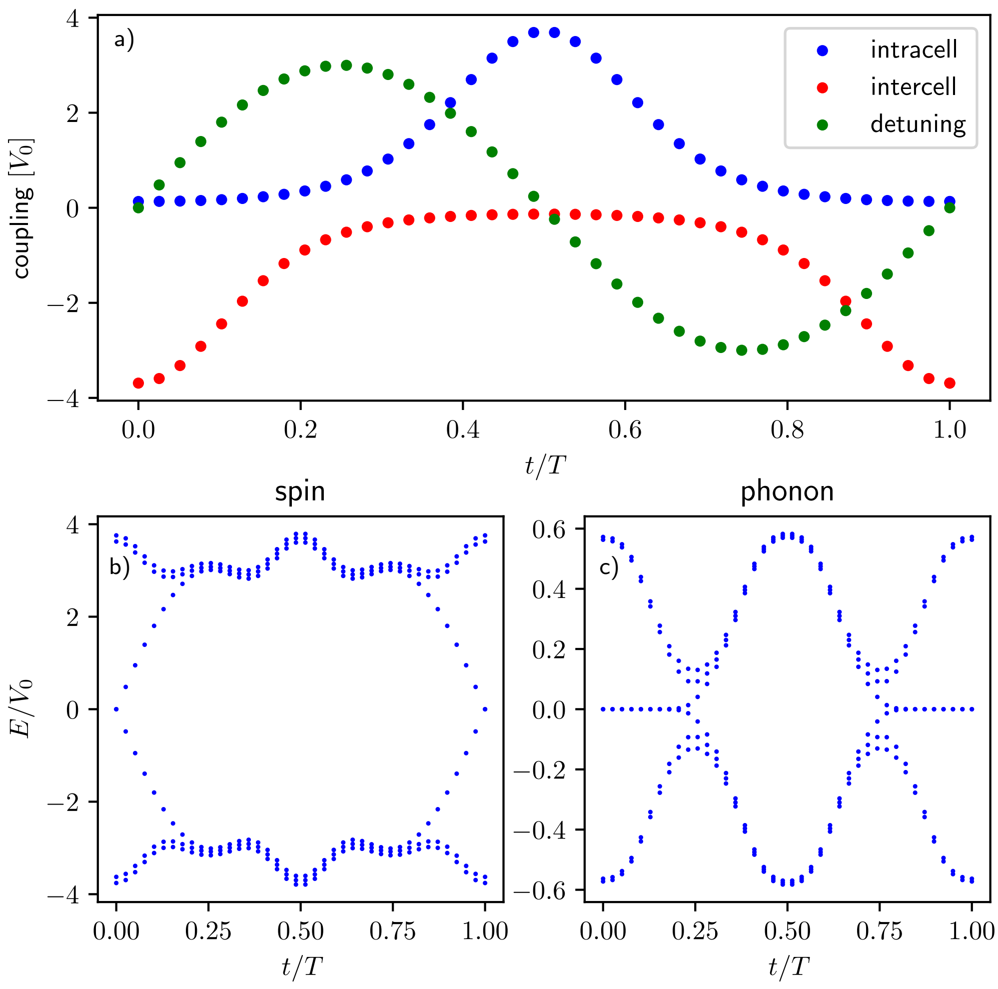

Figure 7: **Spin transfer protocol.** In a the course of the interaction during the transfer protocol is shown. b-c show the spectrum of the spin and phonon Hamiltonian respectively during the pump cycle each for one excitation in the respective degrees of freedom. For the spin Hamiltonian a pure dipole dipole interaction is used, where the dipole moments of the atoms are orientated in way that puts the interactions within the same sublattice to zero. This choice improves the accuracy of the nearest neighbor assumption. The phonon spectrum is received by the methods used in the previous chapters. It therefore depicts a scenario without spin degrees of freedom.

As a next step I want to first discard the phonon dynamics and realize a transfer protocol for a spin excitation through the chain. For this purpose I implement a Rice-Mele model configuration for the chain of Rydberg atoms. This is achieved by including an onsite potential with alternating sign of the form

$$H_{\mathrm{OS}} = \sum_{i=1}^{N} (-1)^i \, \Delta \, \sigma_i^+ \sigma_i^- \, .$$

Note that the detuning Δ is now a different parameter from above. By changing the position and detuning in an adiabatic cyclic way, one can implement a transfer protocol, which is depicted in Fig. 7. The system is initialized with the eigenstate of the spin Hamiltonian located at the edge. This is possible, as I start the transfer protocol in the topological phase. After the the transfer period I compare the evolved state with the target state, which is the eigenstate located at the other side. I use a chain of six atoms in total. The resulting fidelity between final and target state for this protocol is 0.9996.

I can now compare which influence the introduction of photons has on the transfer quality. Therefore I test the effect of four different phonon initial states on the transfer process. The different initial phonon states can be read of Fig. 8. Fig. 9 shows that the evolvement of the phonon degrees of freedom does not change the quality of the transfer to a very considerable portion, where I investigated how the strength of the phonon-phonon interactions effect the fidelity of final and target state. In Fig. 10 I show the time evolution of the phononic degrees of freedom for two initial phonon excitations.

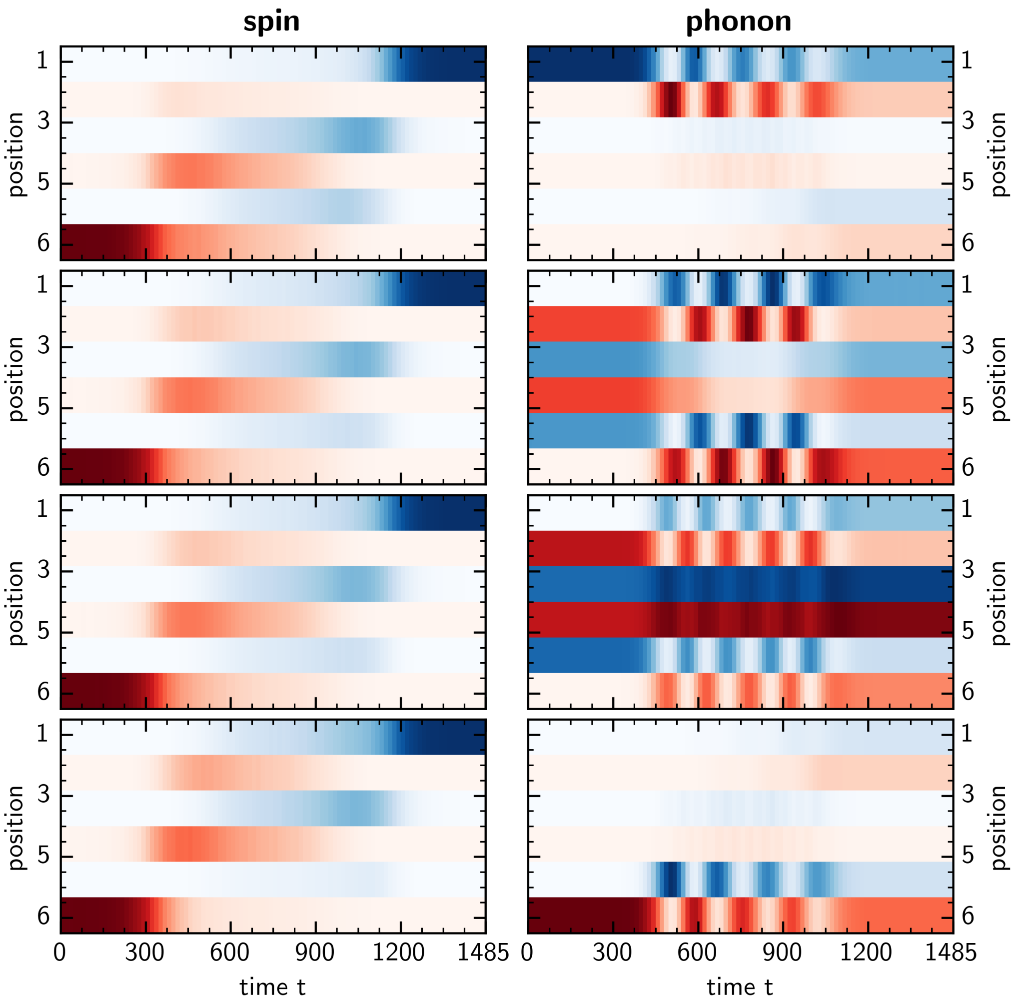

Figure 8: **Spin transfer process.** The time evolution of the spin and phonon degrees of freedom is depicted during the adiabatic transfer protocol. For the same transfer protocol the system was prepared in different initial phonon configurations, which stored one excitation in total.

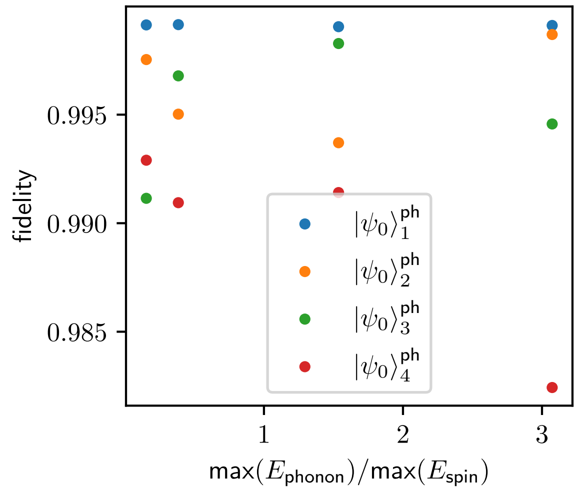

Figure 9: **Fidelity in dependence of phonon interaction strength.** For the 4 initial phonon configurations introduced in Fig. 8 (the labeling runs from top to bottom) different rations of spin dipole interactions to phonon interactions are chosen. The fidelity between the final and target state is shown in dependence of interaction ratio and initial phonon state.

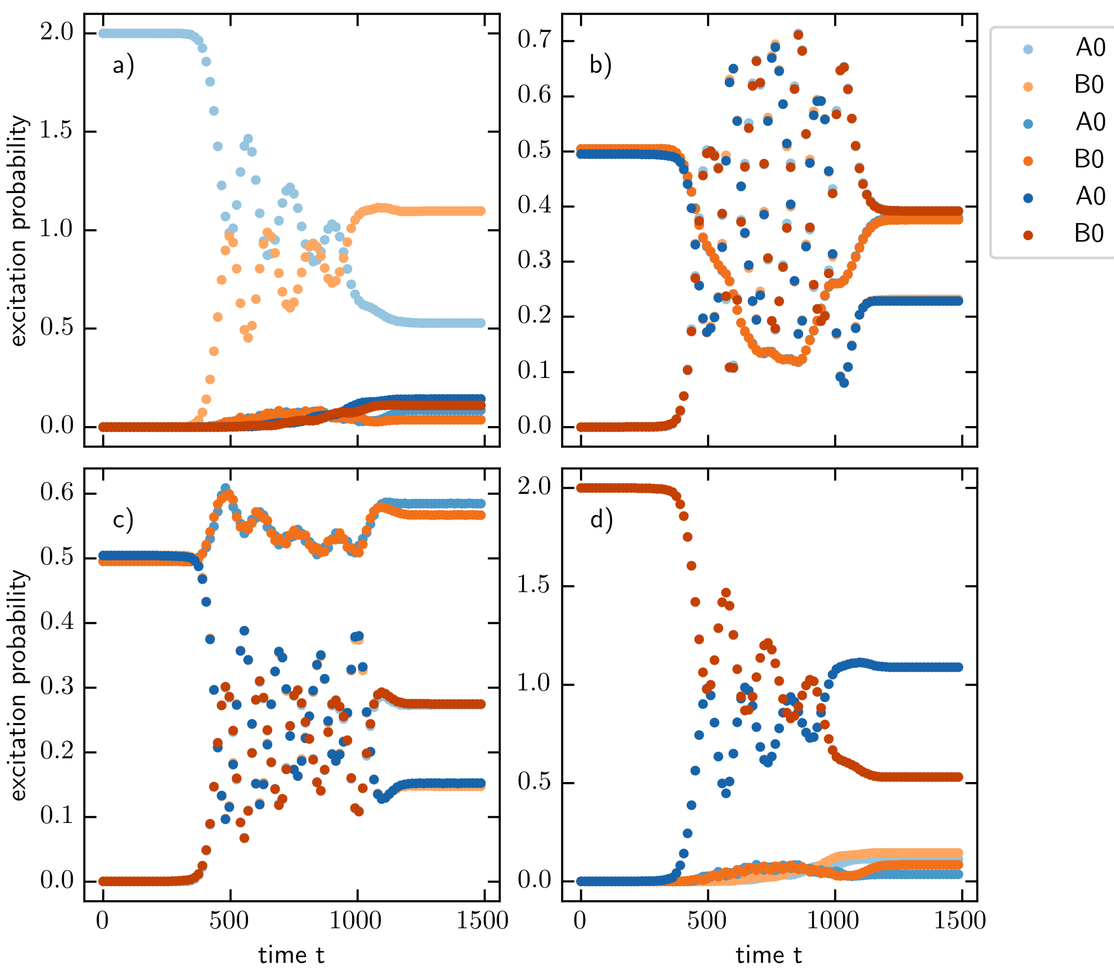

Figure 10: **Phonon evolution for two excitations.** For two phonon excitations the evolution of the phonons during the transfer period is depicted.

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